

Figure 1: The official portal of Machine Learning Database (MLDB)

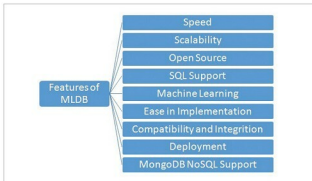


Figure 2: Features of MLDB

is available and distributed on a powerful repository and hosting of GitHub (<https://github.com/mldb-ai/mldb>).

SQL support: This makes MLDB very user-friendly along with the support for Big Data processing. MLDB can process, train and make predictions using database tables that have millions of columns, with concurrent processing and no compromise on integrity.

Machine learning: MLDB is developed for high performance machine learning applications and models. It has support for deep learning with the graphs of TensorFlow that make it superior in knowledge discovery.

Ease of implementation: There are installation packages for multiple platforms and programming environments including Jupyter, Docker, JSON, Cloud, Hadoop, and many others.

Compatibility and integration: MLDB provides a higher degree of compatibility with different application programming interfaces (APIs) and modules including JSON, REST and Python based wrappers.

Deployment: MLDB can be deployed easily on an HTTP endpoint that provides easy interface and fast deployment.

MongoDB and NoSQL support: The bridge or interface of MongoDB and MLDB can be created to support MLDB SQL queries. These SQL queries can be executed on MongoDB collections, which give MLDB more powers to interact with NoSQL databases for unstructured and heterogeneous datasets.

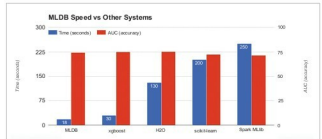


Figure 3: Performance of MLDB compared with other machine learning libraries

Figure 3 depicts the performance of MLDB when compared to other libraries. An execution of the 100 Tree Random Forest approach is done on 1 million rows with one node using MLDB and other libraries. From the graphical results, it is evident that MLDB is comparatively better, takes less time and its accuracy compares well with other machine learning libraries. The performance of MLDB is comparable with that of xgboost, H2O, Scikit-Learn and Spark MLlib.

Support for algorithms in MLDB

The procedures of MLDB are used for the training of machine learning models and these are implemented using functions. Given below is the list of functions and procedures that can be used in MLDB with high performance.

Supervised machine learning

- Classification including multi-label classification: classifier.train
 - Logistic regressions
 - Generalised linear models
 - Neural networks
 - Decision trees
 - Random forests
 - Naive Bayes models using boosting and bagging
- Support vector machines (SVM): svm.train
- Classifiers calibration: probabilizer.train

Deep learning

- TensorFlow models: tensorflow.graph

Clustering

- K-Means models: kmeans.train

Dimensionality reduction, manifold learning and visualisation

- Truncated Singular Value Decompositions (SVD): svd.train
- t-distributed Stochastic Neighbour Embedding (t-SNE): tsne.train

Feature engineering

- SentiWordNet models: import.sentiwordnet
- Word2Vec: import.word2vec
- Term-Frequency/Inverse-Document-Frequency (TF-IDF) models: tfidf.train
- Count-based features: statsTable.train
- Feature Hashing/Vectorize features: feature_hasher